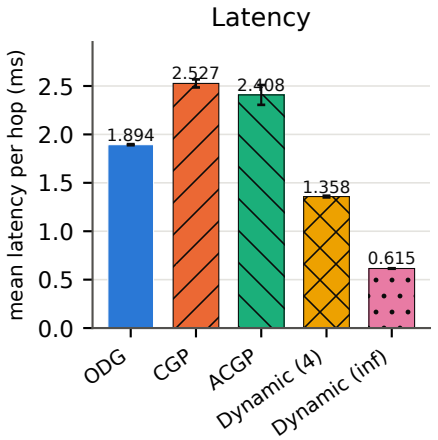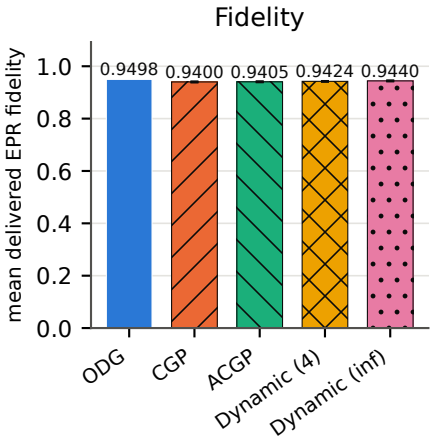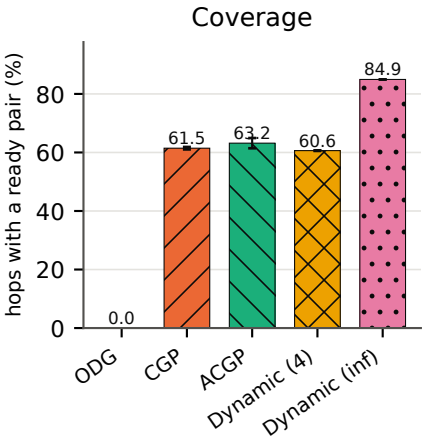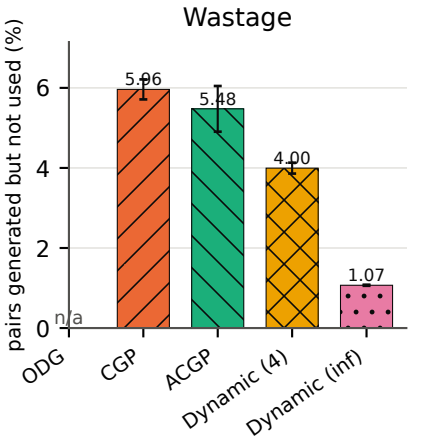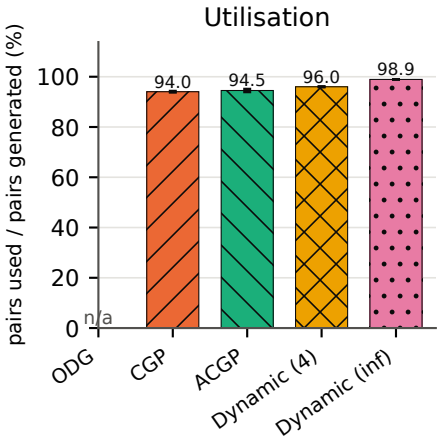QPE (exact), XY routing: 4 memories per core, all 4 for pre-generation